Worksheet 6-9 A Integration with U-Substitution #1
AP Calculus Name_____

**Directions:** Find the following antiderivatives using substitution. Simplify your answers as appropriate

1. $\int 2(2x+1)^4\,dx =$

2. $\int 2x(x^2-9)^3\,dx =$

3. $\int -2x\sqrt{9-x^2}\,dx =$

4. $\int -4x\sqrt[3]{1-2x^2}\,dx =$

5. $\int x^3(x^4+3)^2\,dx =$

6. $\int x(4x^2+3)^3\,dx =$

7. $\int t\sqrt{t^2+2}\,dt =$

8. $\int u^2\sqrt{u^3+2}\,du =$

9. $\int \frac{x^2}{(1-x^3)^2}\,dx =$

10. $\int \frac{x}{\sqrt{1-x^2}}\,dx =$

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11. $\int \left(\frac{x^3}{\sqrt[3]{1+x^4}}\right) dx =$

12. $\int \frac{2}{\sqrt[3]{2x-5}} dx =$

13. $\int \left(1+\frac{1}{t}\right)^3 \left(\frac{1}{t^2}\right) dt =$

14. $\int \frac{3}{2x-1} dx =$

15. $\int \frac{2x}{x^2+7} dx =$

16. $\int 3e^{4x+5}\, dx =$

17. $\int 2xe^{3x^2-2}\, dx =$

18. $\int \sin(2\theta)\, d\theta =$

19. $\int 2x \sec^2(x^2+5)\ dx =$

20. $\int 3x \cos(x^2)\ dx =$

21. $\int \sec(1-t) \tan(1-t)\, dt =$

22. $\int 5x^3 \sin(x^4-1)\, dx =$